

Quiz-2
40 minutes

Basics of Electrical Engineering (EE100)
20 marks

1. The dimensions of the core of a magnetic system excited by a single coil of 200 turns is as follows: The mean length of the core is 150 cm and the area of cross section is 100 cm^2 . Given that the relative permeability of the core, $\mu_r = 1000$; permeability of free space, $\mu_0 = 4\pi \times 10^{-7} \text{ H/m}$; and the steady current flowing through the coil, $i = 2 \text{ A}$.

- (a) Determine the reluctance of the core
- (b) Determine the flux in the core.
- (c) Determine the self-inductance of the coil

[10]

2. A balanced delta connected load has a phase current, $I_{AB} = 10 \angle 30^\circ \text{ A}$ and is supplied by a star connected source with line voltage, $V_{AB} = 100 \angle 0^\circ \text{ V}$. Assume abc phase sequence and balanced supply. The phasors are expressed in rms values.

- (a) Draw the circuit diagram of the system
- (b) Determine line current, I_A and phase current I_{CA} . Draw phasor diagram showing these currents.
- (c) Determine power consumed in the load
- (d) Determine the power factor of the load

[10]

Quiz -2

Each question carries 10 marks. Exam is for 20 marks and 40 minutes

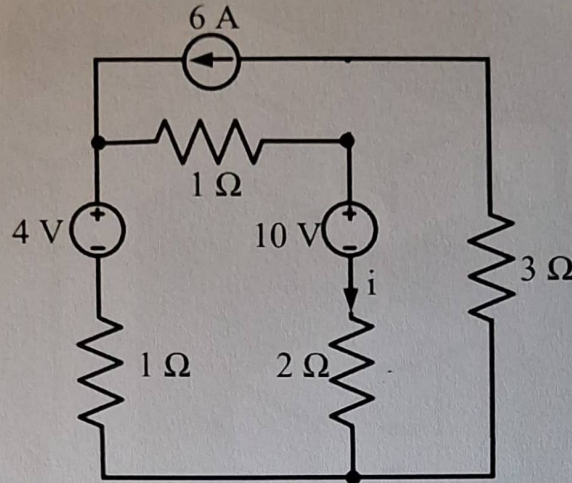
1. The RMS values of the voltages measured across the resistor R and the inductor L in a series RL circuit supplied by an AC voltage source v_s are 160 V and 120 V respectively. The RMS value of the current drawn from the supply is 10 A.
 - a. Determine the RMS value of the voltage applied by the source
 - b. Draw the phasor diagram showing the all voltages and the currents
 - c. Determine the value of R and L , if the angular frequency ω of the voltage source is 1000 rad/s
 - d. Determine the value of the capacitor that must be connected in series with the RL load so that the overall series combination behaves as a pure resistor.
2. A 200 V, 50 Hz ac voltage source supplies two loads which are connected in parallel, whose details are given below:

Load-1: 4 kW, 0.8 pf lag
Load-2: $Z_2 = 20 - 20j \Omega$
Determine:
 - (a) ^{real} Power and reactive power supplied by the source
 - (b) Current supplied by the source
 - (c) Value of capacitor required to be connected in parallel with the loads to make the power factor unity

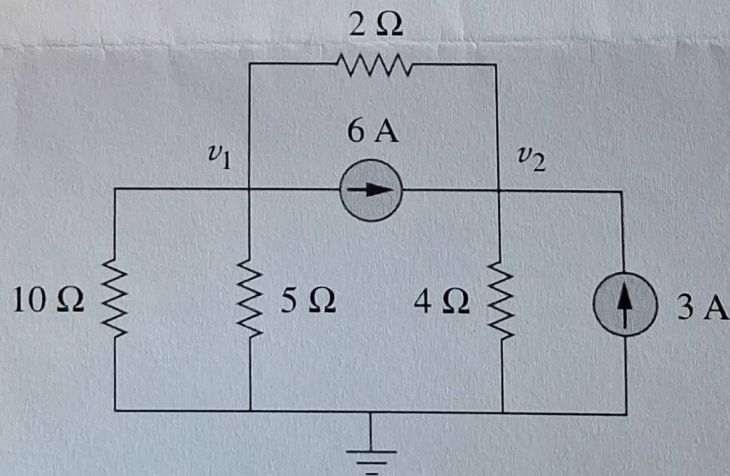
Quiz -1

Each question carries 10 marks. Exam is for 20 marks and 40 minutes

1. Use mesh analysis to determine (a) current, i and (b) power supplied by 10 V source in the circuit shown. Don't solve it using other methods.



2. The linear electrical circuit shown in the figure can be modelled as: $\mathbf{A} \mathbf{x} = \mathbf{b}$, where $\mathbf{x}$ is a column vector of two entries given by: $\mathbf{x} = [v_1, v_2]^T$. Find $\mathbf{A}$ and $\mathbf{b}$



Name:

Roll no.

Quiz -2

20 marks

40 minute

Answer in the question paper itself. If needed, use additional sheet.

A 500 V, 50 Hz, balanced three-phase delta-connected source supplies power to two balanced three-phase loads, L_1 and L_2 , connected in parallel.

L_1 : Balanced delta-connected load with impedance $(40+j30) \Omega$ per phase

L_2 : Balanced star-connected load with impedance $(40+j30) \Omega$ per phase

(a) Draw the circuit diagram of the system.

(b) Given that the line voltage $V_{AB} = 500 \angle 0^\circ$ V, determine:

the phase current, I_{BC} and the line current, I_A drawn by the delta-connected load L_1 .

(c) Calculate the real power consumed by the delta-connected load L_1 .



Roll No 25CSE1022

राष्ट्रीय प्रौद्योगिकी संस्थान गोवा
NATIONAL INSTITUTE OF TECHNOLOGY GOA

Cuncolim, South Goa, 403703

Programme Name: B.Tech.

Mid Semester Examinations, October -2025

Course Name: Basics of Electrical Engineering

Date: 13/10/2025

Duration: 1.5 Hours

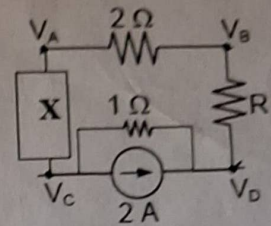
Course Code: EE 100

Time: 02:00 pm - 03:30 pm

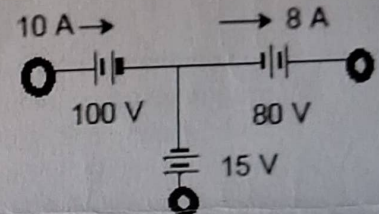
Max. Marks: 50

ANSWER ALL QUESTIONS

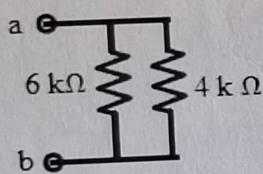
1. Given that X is an electrical network, whose details are not known
- Find $V_C - V_D$, if $V_A - V_B = 6\text{ V}$.
 - Find the value of R, if the power dissipated in R is half of the power dissipated in 2Ω resistor [4M]



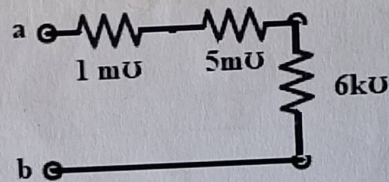
2. The three batteries shown in the figure part of an electrical circuit.
- Determine the power supplied by each battery
 - Which of the batteries are getting charged
- [4M]



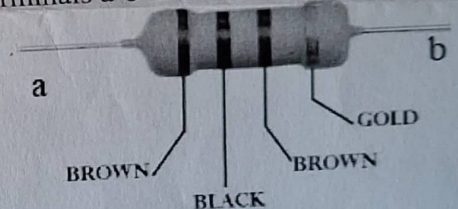
3. Find the equivalent resistance of the network as seen from terminals a-b in each case



(a)



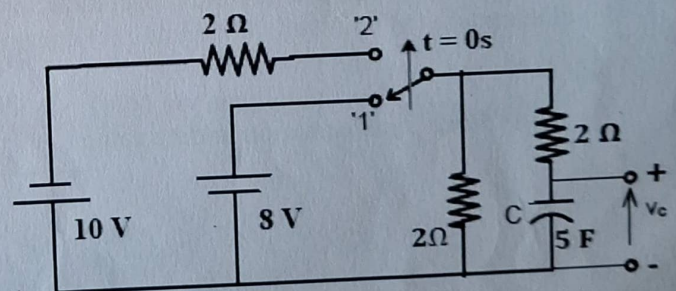
(b)



(c)

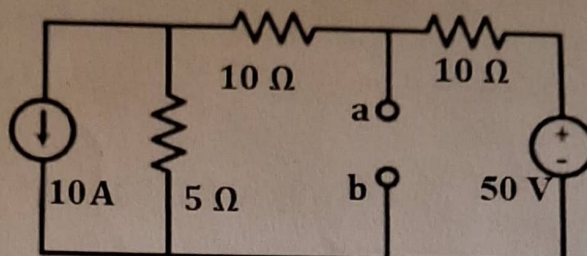
[3x2=6 M]

4. The switch given in the Figure was in position 1 for a long time, and is moved to position 2 at time, $t = 0\text{ s}$. Determine the value of:
- $v_C(0)$
 - $v_C(\infty)$
 - $v_C(20\text{ s})$
 - $v_C(t)$, for all $t > 0$
- [8 M]



5. (a) Find v_{ab} using superposition theorem
 (b) Find the Norton equivalent of the network as seen from terminals a-b.
 (c) Find the maximum power that can be transferred to a resistance connected between terminals a and b

[12 M]



6. Consider a two-terminal device whose v - i relationship is given by the following expression:
 (a) $v = 5 - 2i$ (b) $v = 3i + 2 \frac{di}{dt}$

Draw a two-terminal circuit (using any standard elements) that reproduces the specified v - i characteristic for each case. You may use more than one element for a given case. [4 M]

7. The reading of a house hold energy meter was 1287.5 kWh at the beginning of the month. The equipment usage details for the house hold are given in the table below. (a) What will be the reading at the end of the month? (b) What will be electricity charges for the month, if the cost of electricity is Rs 5.5/ unit?

[4M]

Equipment	Wattage(W)	No. of equipment	No of hours/ day/equipment
Bulb	20	10	6
Fan	40	3	8
Refrigerator	300	1	8
AC	1100	1	4

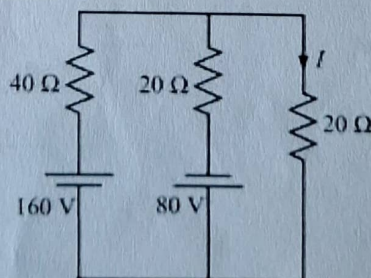
8. The voltage, v measured across an inductor of 2 mH is given by: $v = \begin{cases} 1 \text{ V}, 0 \leq t < 2 \text{ ms} \\ -2 \text{ V}, 2 \text{ ms} \leq t < 3 \text{ ms} \end{cases}$
 Given that the initial current through inductor, $i(0) = 5 \text{ A}$.

Determine (a) $i(2 \text{ ms})$, (b) $i(3 \text{ ms})$, (c) energy store in inductor at $t = 3 \text{ ms}$.

[4M]

9. Use source transformation to determine I

[4M]





Roll No 25C5E1049

राष्ट्रीय प्रौद्योगिकी संस्थान गोवा
NATIONAL INSTITUTE OF TECHNOLOGY GOA

Cuncolim, South Goa, 403703

Programme Name: B.Tech.

Mid Semester Examinations, March -2026

Course Name: Basics of Electrical Engineering

Date: 07/03/2026

Duration: 1.5 Hours

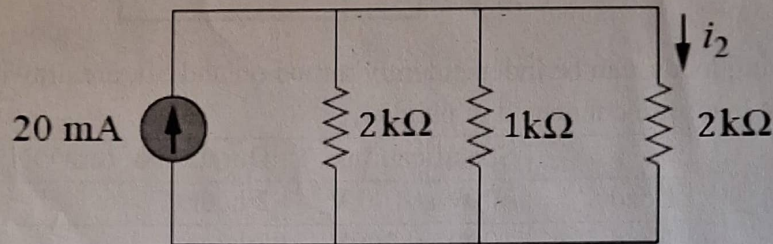
Course Code: EE 100

Time: 10:00 am - 11:30 am

Max. Marks: 50

ANSWER ALL QUESTIONS

1. For the circuit given, determine (a) i_2 , (b) power supplied by 20 mA source [6]

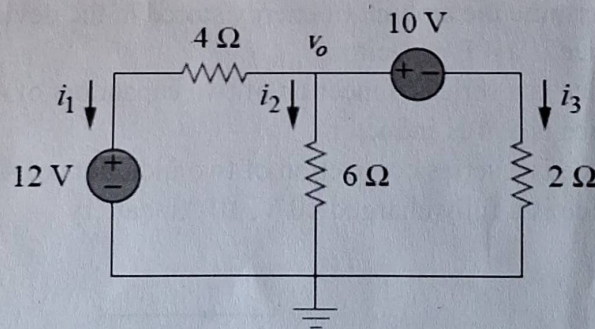


2. Create a circuit (draw the circuit diagram) which can be modelled as:

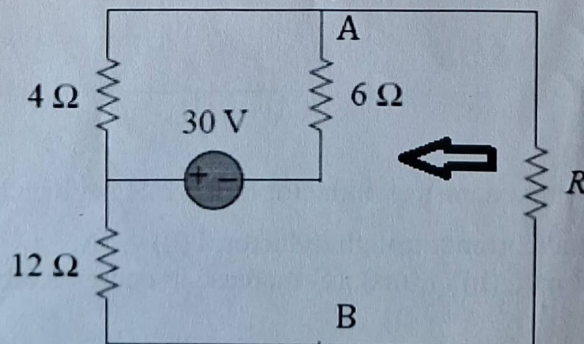
$$\begin{bmatrix} 2 & -1 \\ -1 & 4 \end{bmatrix} \begin{bmatrix} i_1 \\ i_2 \end{bmatrix} = \begin{bmatrix} 3 \\ 2 \end{bmatrix}, \text{ where } i_1 \text{ and } i_2 \text{ are the mesh currents.}$$

[6]

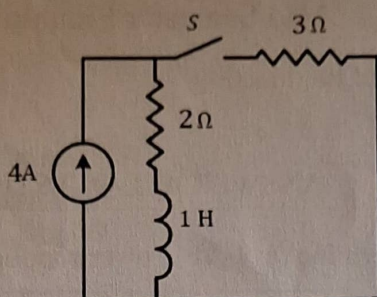
3. What is super position theorem? Using superposition theorem, determine v_o [7]



4. Find the Thevenin equivalent of the part of the circuit which is on the left hand side of terminals AB. What is the maximum power that can be supplied to the variable external resistance R ? [7]



5. The switch shown in the figure was in open position for a very long amount of time. At time $t=0$, the switch was closed. (a) Determine the time constant of the new circuit which is formed after $t=0$. (b) Find the voltage across the 2Ω resistor as a function of time, for $t > 0$. [6]

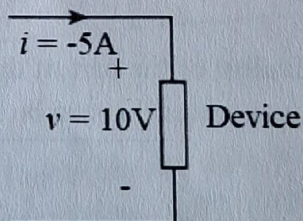


6. The following loads, can be independently turned on and off, are drawing power from a 100 V supply. The loads are connected in parallel.

	Specifications	Duration of turn ON
Load-1	40 W at 100 V	2 hours
Load-2	100W at 200V	3 hours
Load-3	50 Ω	2 hours

- (a) Determine the power drawn by each load from 100 V source
 (b) Determine the energy consumed by each load
 (c) Find the total cost of electricity used by these loads, if one unit costs Rs 6.00. [7]

7. The voltage across and the current flowing through the device shown in Figure is 10 V and -5A respectively. Determine the amount of energy stored in the device, if
- the given device is a 4 F capacitor
 - the given device is a series connection of two capacitors of 4 F each.
 - the given device is a 4 H inductor
 - the given device is a series connection of two inductors of 4H each.
 - the given device is a fully charged 10 V, 10 Ah battery
- [5]



8. The voltage, v measured across an inductor of 0.5 mH is given by: $v = \begin{cases} 1 \text{ V}, & 0 \leq t < 2 \text{ ms} \\ -2 \text{ V}, & 2 \text{ ms} \leq t < 3 \text{ ms} \end{cases}$
 Given that the initial current through inductor, $i(0) = 5\text{A}$.
 Determine (a) $i(2 \text{ ms})$, (b) $i(3\text{ms})$, (c) increase in energy stored in the inductor during this 3ms time. [6]



Roll No	25	C	S	E	17	0	2	2
---------	----	---	---	---	----	---	---	---

राष्ट्रीय प्रौद्योगिकी संस्थान गोवा
NATIONAL INSTITUTE OF TECHNOLOGY GOA
Cuncolim, Salcete, South Goa -403703
Programme Name: B.Tech.
End Semester Examinations, December-2025

Course Name: Basics of Electrical Engineering
Date: 16.12.2024
Duration: 3 hour

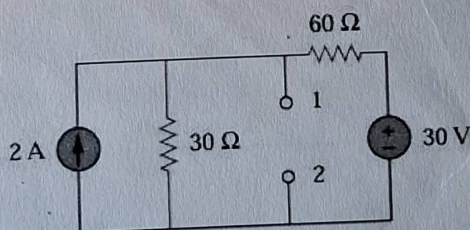
Course Code: EE 100
Time: 10:00 am-01:00 pm
Max. Marks: 100

ANSWER ALL QUESTIONS

Part-A is to be answered **first** and is to be returned within **one hour** of the exam.
Part-A has to be answered in the **question paper** itself.
Part-B has 11 questions and carries a total of 60 marks

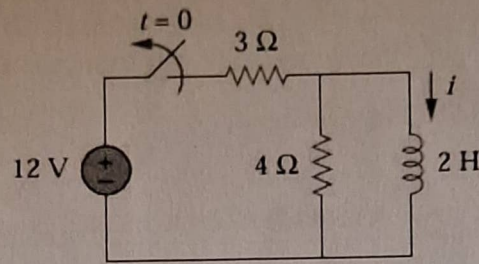
Part-B

1. With the help of a neat diagram, briefly explain the parts and operation of a hydro-electric power plant. [5]
2. In a power transformer, the applied voltage and frequency remain constant during operation, whereas the current varies with load. Identify the different losses that occur in a transformer. Which of these losses vary with load? Explain the constructional features adopted to minimize these losses. [5]
3. Derive the expression for the instantaneous power, $p(t)$ drawn by a linear load of 1Ω when supplied from an ac source, $v_s(t) = 10 \cos(100t)$. Hence, show that the instantaneous power is a periodic waveform with an angular frequency of 200 rad/s. [6]
4. Using the superposition theorem, demonstrate how to determine the Thevenin equivalent of the given circuit as seen from terminals 1 and 2. [6]



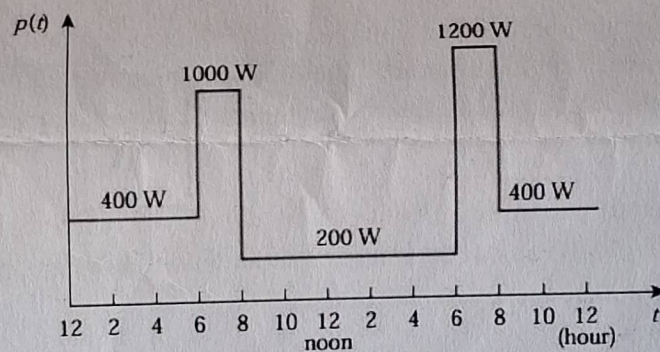
5. Design a series RLC tuning circuit. The minimum value of impedance offered by the network should be equal to 5Ω at $\omega = 1$ rad/sec. The magnitude of the impedance should be equal to 13Ω when the applied frequency becomes $\omega = 2$ rad/s. (a) Determine the value of R, L and C used. (b) What is the value of impedance offered by this network if the applied signal is dc? [5]

6. The switch shown in the figure was in ^{closed} open position for a very long duration of time (from $t = -\infty$). At time $t = 0$, the switch is open

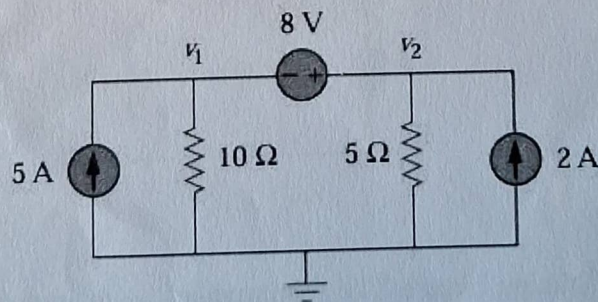


Determine: (a) i_0^- , (b) i_0^+ , (c) i_∞ (d) Time constant of the circuit for $t \geq 0$, (e) i_t for $t \geq 0$. [6]

7. The power consumption of a house hold for a day is shown in the Figure. Find (a) energy consumed in a day, (b) average power during the day, (c) amount to be paid as electricity charges for the day if electricity costs Rs 10 per unit. [6]



8. Find v_1 and v_2 using mesh analysis. [6]



9. The given tables provide information about the Indian electricity sector, showing data on the installed electricity capacity by various sources as of the year 2024, and electricity generation by source in the years 2019 and 2024.

Sl. No.	Energy Source	Installed capacity (GW)	Electricity generation (TWh)	
		2024	2019	2024
1	Coal	210	1010	1180
2	Natural Gas	25	50	45
3	Nuclear	7	40	48
4	Large Hydro	47	150	160
5	Solar	82	50	140
6	Wind	45	60	95
7	Biomass	11	30	35
	Total	432 GW	1390 TWh	1703 TWh

The estimated population of India in 2019 and 2024 is 1.37 billion and 1.44 billion, respectively.

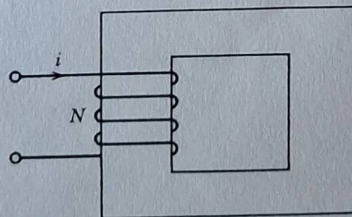
- What percentage of electricity generation was contributed by fossil fuels in the year 2024?
- Which energy source showed the highest and lowest percentage growth in electricity generation between 2019 and 2024? Calculate the respective growth rates (%).
- Determine the overall capacity factor of coal-based power plants in the year 2024.
- What percentage of the total electricity generated in 2024 came from new renewable energy sources? (Exclude large hydro, as it is considered a conventional source.)
- Calculate the per-capita electricity consumption in India for the year 2024.
- Which is growing at a faster rate: population or per-capita electricity consumption? Calculate and compare the growth rates of both.

[10]

10. The mean length and the area of the core given in Figure is 200 cm and 80 cm² respectively. Given that the relative permeability of the core, $\mu_r = 2000$; permeability of free space, $\mu_0 = 4\pi \times 10^{-7}$ H/m; number of turns, $N=200$; and the steady current flowing through the coil, $i = 3$ A.

- Determine the reluctance of the core
- Determine the flux in the core
- Determine the self-inductance of the coil
- Draw λ vs i (flux linkage vs current) curve for the given coil. Is it linear?

[5]





Roll No 25 C S E 1 0 4 9

राष्ट्रीय प्रौद्योगिकी संस्थान गोवा
NATIONAL INSTITUTE OF TECHNOLOGY GOA
Cuncolim, Salcete, South Goa -403703
Programme Name: B.Tech.
End Semester Examinations, April-2026

Course Name: Basics of Electrical Engineering
Date: 06.05.2026
Duration: 3 hour

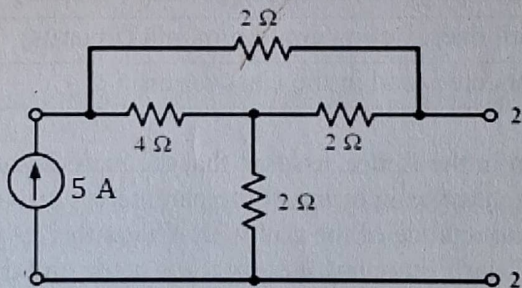
Course Code: EE 100
Time: 10:00 am-01:00 pm
Max. Marks: 100

ANSWER ALL QUESTIONS

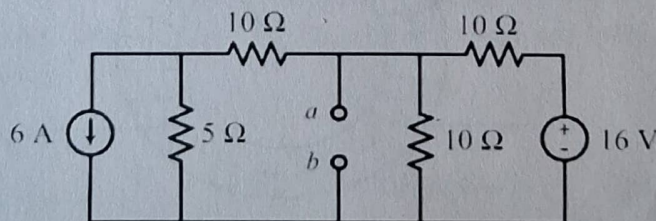
Question numbers 1- 10 carries 6 marks each

[10 x 6 = 60]

1. Name any three types of passive components used in a linear electrical network. For each component, express its characteristic relationship in the form of an algebraic equation $y = m x$. Define the variables x and y in each case. Also, identify the physical quantity whose dimension is the same as that of the product xy .
2. A battery has an electromotive force of 10 V and an internal resistance of 1 m Ω . Calculate the power supplied by the battery to each of the following load conditions: (a) 1 k Ω resistor (b) parallel combination of 1k Ω and 1 Ω resistors, (c) 1m Ω resistor
3. With the help of neat, labeled diagrams (such as a cross-sectional view), describe the constructional details of the iron core used in a core-type transformer. What types of losses occur in the transformer core? Explain how these losses are minimized or reduced during the construction of the core.
4. Find the Thevenin equivalent of the network as seen between terminals, 2-2'



5. Find v_{ab} using superposition theorem.



6. Design a series RLC tuning circuit. The minimum value of impedance offered by the network should be equal to 10 Ω at $\omega = 1$ rad/sec. The magnitude of the impedance should be equal to 13 Ω when the applied frequency becomes $\omega = 2$ rad/s. (a) Determine the value of R, L and C used. (b) What is the value of impedance offered by this network if the applied signal is dc?

7. The table below provides details of the Indian electricity sector, including installed capacity as of 2023 and electricity generation for the years 2018 and 2023:

Source	Installed capacity (GW) in 2023	Electricity produced (TWh) in the year 2018	Electricity produced (TWh) in the year 2023
Coal+lignite	211.8	1022	1294
Natural gas+oil	24.8	50	32
Large Hydro	46.9	134	134
Wind	42.6	62	83
Solar	66.8	39	116
Other renewables	20.6	25	25
Nuclear	6.7	38	48
Total	416	1371	1734

(a) Calculate the percentage of electricity generated from fossil fuels for the given years. Which year has a higher percentage?

(b) Determine the average capacity factor of large hydro power plants in 2023. 0.3261539

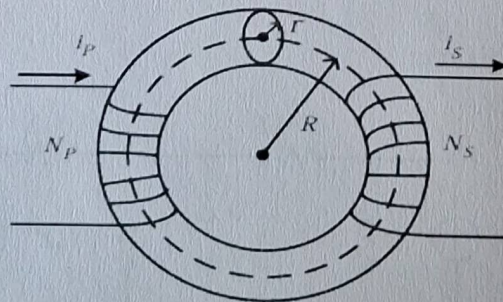
(c) Calculate the percentage of total electricity generated in 2023 that comes from new and renewable energy sources (exclude large hydro, as it is considered a conventional source). 12.91

8. Make a rough estimate of the monthly energy consumption in your class room in the given format:

Devices	No	Approx. power rating/device	Average no of hours of operation/day	Average energy consumption/day
LED bulbs		25 W		
Fans		40W		
Projector		300 W		
Estimated total energy consumption per day (in units)				
Estimated total energy consumption/month (in units)				
Total connected load in the class room(in W)				

9. For the magnetic core shown in the figure, assume that the material has a relative permeability μ_r . Derive expressions for the following quantities in terms of the parameters provided in the figure:

(a) reluctance of the core, (b) inductance of the coil with N_p number of turns, (c) mutual inductance between the coils shown in the figure. Clearly state any assumptions made and define all symbols used.



10. A capacitor of capacitance $0.1 \mu\text{F}$, initially charged to 100V , is discharged through a resistor of resistance $1 \text{ k}\Omega$. Determine:

a) The time constant of the circuit.

b) The time required for the voltage across the capacitor to decrease to 1V .

c) The amount of heat energy dissipated in the resistor during this interval.

11. The signal v is a periodic square waveform of time period 2s, and signal u is sinusoidal.

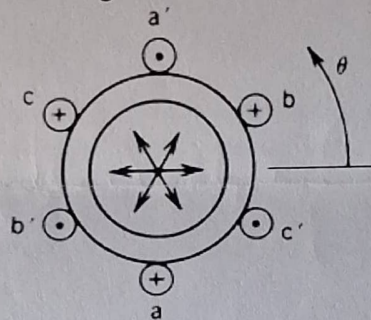
$$v = \begin{cases} -10 \text{ V}, & 0 < t < 1 \text{ s} \\ 10 \text{ V}, & 1 < t < 2 \text{ s} \end{cases} \quad \text{and } u = 10 \sin(2t - 45^\circ). \text{ Complete the table with appropriate values.}$$

Waveform	Frequency (Hz)	Peak value (V)	Rms value (V)	average value (V)
v				
u				

12. The voltage across and current through a load is given as: $v = 10 \cos(2t - 45^\circ)$ and $i = 4 \cos(2t - 60^\circ)$.

- Find the expression for the instantaneous power, p drawn by the load
- Plot the following waveforms for $0 < \omega t < 2\pi$, (a) v vs ωt , (b) i vs ωt , (c) p vs ωt
- Find the frequency of p , if it is a periodic waveform.
- Determine: (a) real power, P (b) power factor, (c) complex power, S

13. A cross-sectional view of a three-phase induction machine, along with its three-phase stator windings, is shown in the figure. (a) Identify and mark the magnetic axes of the three stator windings on the diagram. (b) With the help of appropriate diagrams, explain how a rotating magnetic field of constant magnitude is produced when three-phase currents flow through the stator windings.



14. A balanced delta connected load draws a line current of 10 A at **unity power factor** from a balanced three phase source of line-to-line voltage of 100 V at 50 Hz.

- Draw the circuit diagram of the system
- Draw the phasor diagram showing line voltage V_{ab} , line current I_a , phase current I_{ab} .
- Determine the impedance of each phase of the load.
- Calculate the real and reactive power consumed by the load

15. The elements shown in the figures represent parts of different circuits. Determine the (a) total power absorbed by the three elements in circuit A, (a) total power dissipated by three elements in circuit B, given that the power dissipated by the 4Ω resistor is 40W, (c) total energy stored in the three elements in circuit C, given that the energy stored in C_1 is 10 J.

